

Fibonacci Numbers

Neil Atreya

PROBLEM 1.1.

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597

Therefore, the answer is 1597

PROBLEM 1.3.

Looking at the fibonacci numbers:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...

The even numbers are:

$F_0 = 0, F_3 = 2, F_6 = 8, F_9 = 34, F_{12} = 144, \dots$

Therefore, by observation:

That is why F_{n+3} is the function to get the next even number, if F_n is an even number.

Why this happens is because:

- every even + odd number equals an odd number
- every even + even number is another even number
- every odd + odd number is another even number

When you're going in order for this:

1. $0 + 1$: even + odd = odd.
2. $1 + 1$: odd + odd = even.
3. $1 + 2$: odd + even = odd.

That's why every third number of n is the function to get the next even number

PROBLEM 1.4.

Looking at the fibonacci numbers:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...

The multiples of 3 are:

$F_4 = 3, F_8 = 21, F_{12} = 144, \dots$

Therefore, by observation:

If F_n is a multiple of 3, then F_{n+4} is the function to get the next multiple of 3.

Why?:

$$F_{n+4} = F_{n+3} + F_{n+2}$$

$$F_{n+3} = F_{n+2} + F_{n+1}$$

$$F_{n+2} = F_{n+1} + F_n$$

$$F_{n+4} = F_{n+2} + F_{n+1} + F_{n+1} + F_n$$

$$F_{n+4} = F_{n+1} + F_n + F_{n+1} + F_{n+1} + F_n$$

$$F_{n+4} = 3(F_{n+1}) + 2(F_n)$$

$F_n = 3k$, since F_n is a multiple of 3.

$$F_{n+4} = 3(F_{n+1}) + 6k$$

$$F_{n+4} = 3(F_{n+1} + 2k)$$

This shows F_{n+4} is a multiple of 3.

That is why F_{n+4} is the function to get the next multiple of 3, if F_n is a multiple of 3.

PROBLEM 1.5.

```
fib_prev = 0
fib_curr = 1

print(fib_prev)
print(fib_curr)

for i in range (2, 100):
    fib_next = fib_prev + fib_curr
    print (fib_next)
    fib_prev = fib_curr
    fib_curr = fib_next
```

This code prints the first 100 Fibonacci numbers. In problems 1.3 and 1.4, the rule is that F_{n+3} is the function for every next even number. And F_{n+4} is the function for every next multiple of 3.

TODO

That means that for every number m, there is always going to be an infinite umber of multiples of m's in the Fibonacci sequence.

PROBLEM 1.9.

Compute fibonacci ratios:

$$\frac{F_{n+1}}{F_n}$$

For $1 \leq n \leq 10$:

$$1, 2, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \frac{13}{8}, \frac{21}{13}, \frac{34}{21}, \frac{55}{34}, \frac{89}{55}$$

Approximately:

$$1, 2, 1.5, 1.667, 1.6, 1.625, 1.615, 1.619, 1.618, 1.618$$

So yes, they seem to approach the **golden ratio** 1.618...

Compute Lucas ratios:

$$\frac{L_{n+1}}{L_n}$$

For $1 \leq n \leq 10$:

$$\frac{3}{1}, \frac{4}{3}, \frac{7}{4}, \frac{11}{7}, \frac{18}{11}, \frac{29}{18}, \frac{47}{29}, \frac{76}{47}, \frac{123}{76}, \frac{199}{123}$$

Approximately:

$$3, 1.333, 1.75, 1.571, 1.636, 1.611, 1.621, 1.617, 1.618, 1.618$$

So yes, they seem to approach the **golden ratio** 1.618...

PROBLEM 1.10.

To solve $G_n = F_1 + F_2 + F_3 + \dots + F_n$ You first have to start with

$$\begin{aligned} F_1 &= \cancel{F_3} - F_2 \\ F_2 &= \cancel{F_4} - \cancel{F_3} \\ F_3 &= \cancel{F_5} - \cancel{F_4} \\ &\dots \\ F_{n-1} &= \cancel{F_{n+1}} - \cancel{F_n} \\ F_n &= F_{n+2} - \cancel{F_{n+1}} \end{aligned}$$

Everything except two terms cancel out!

$$\begin{aligned} G_n &= F_{n+2} - F_2 \\ G_n &= F_{n+2} - 1 \end{aligned}$$

Thus $G_n = F_{n+2} - 1$ is the answer.

To solve $M_n = L_1 + L_2 + L_3 + \dots + L_n$ You first have to start with

$$\begin{aligned} L_1 &= \cancel{L_3} - L_2 \\ L_2 &= \cancel{L_4} - \cancel{L_3} \\ L_3 &= \cancel{L_5} - \cancel{L_4} \\ &\dots \\ L_{n-1} &= \cancel{L_{n+1}} - \cancel{L_n} \\ L_n &= L_{n+2} - \cancel{L_{n+1}} \end{aligned}$$

Everything except two terms cancel out!

$$\begin{aligned}M_n &= L_{n+2} - L_2 \\M_n &= L_{n+2} - 3\end{aligned}$$

Thus, $\boxed{M_n = L_{n+2} - 3}$ is the answer.

PROBLEM 1.11.

To solve $H_n = F_1 + F_3 + F_5 + \dots + F_{2n-3} + F_{2n-1}$ You first have to start with

$$\begin{aligned}F_1 &= \cancel{F_2} - F_0 \\F_3 &= \cancel{F_4} - \cancel{F_2} \\F_5 &= \cancel{F_6} - \cancel{F_4} \\&\dots \\F_{2n-3} &= \cancel{F_{2n-2}} - \cancel{F_{2n-4}} \\F_{2n-1} &= F_{2n} - \cancel{F_{2n-2}}\end{aligned}$$

Everything except two terms cancel out!

$$\begin{aligned}H_n &= F_{2n} - F_0 \\H_n &= F_{2n} - 0 = F_{2n}\end{aligned}$$

Thus, $\boxed{H_n = F_{2n}}$ is the answer.

PROBLEM 1.12.

Computing by example:

$$n = 1 : 1^2 = 1$$

$$n = 2 : 1^2 + 1^2 = 2$$

$$n = 3 : 1^2 + 1^2 + 2^2 = 6$$

$$n = 4 : 1^2 + 1^2 + 2^2 + 3^2 = 15$$

$$n = 5 : 1^2 + 1^2 + 2^2 + 3^2 + 5^2 = 40$$

So the sums are, 1, 2, 6, 15, 40, ...

By observation and trial, products of two successive Fibonacci numbers are:

$$F_1 F_2 = 1 \cdot 1 = 1$$

$$F_2 F_3 = 1 \cdot 2 = 2$$

$$F_3 F_4 = 2 \cdot 3 = 6$$

$$F_4 F_5 = 3 \cdot 5 = 15$$

$$F_5 F_6 = 5 \cdot 8 = 40$$

Based on this pattern:

$$\boxed{F_1^2 + F_2^2 + \dots + F_n^2 = F_n F_{n+1}}$$

PROBLEM 1.13.

Computing by example:

$$n = 2 : F_1 F_3 - F_2^2 = (1)(2) - 1^2 = 1$$

$$n = 3 : F_2 F_4 - F_3^2 = (1)(3) - 2^2 = -1$$

$$n = 4 : F_3 F_5 - F_4^2 = (2)(5) - 3^2 = 1$$

$$n = 5 : F_4 F_6 - F_5^2 = (3)(8) - 5^2 = -1$$

So it alternates: 1, -1, 1, -1, ...

Based on this pattern:

$$\boxed{F_{n-1} F_{n+1} - F_n^2 = (-1)^n}$$